ShopSmart Retail

Business Understanding Report

Project: Week 2 — Data Preprocessing & Feature Engineering for Machine Learning

1. Introduction

ShopSmart Retail is a retail dataset analysis project designed to prepare transactional, customer, product, and store data for machine learning. The project focuses on understanding purchasing patterns and return behavior while preparing reliable data for predictive modelling.

The dataset, master_dataset.csv, contains 50,000 records and 37 original columns. These columns cover order information, customer demographics, product characteristics, store locations, sales amounts, and return status.

2. Business Objectives

The primary objective is to prepare the ShopSmart Retail data for two predictive modelling tasks:

1. **Sales prediction (regression):** Predict Order_Total using relevant information about orders, products, customers, and transactions.
2. **Return prediction (classification):** Predict Return_Flag, where 1 indicates a returned record and 0 indicates a non-returned record.

These tasks can help a retail business understand factors associated with transaction values and product returns. They may also support better planning, customer service, and operational decision-making when the resulting models are properly evaluated.

3. Business Questions

The analysis explores the following questions:

- How are order totals distributed across transactions?
- Which product categories account for the largest numbers of records?
- How does order activity vary by month?
- Which numerical features have strong relationships with order totals?
- Do order-total distributions differ across customer segments?
- Do return rates vary across product categories?

4. Key Findings from Exploratory Analysis

4.1 Order-total distribution

The histogram shows that order totals are concentrated more heavily among lower-value transactions, with fewer transactions at higher values. This suggests a right-skewed distribution and substantial variation in transaction values.

4.2 Product-category distribution

All five product categories contain more than 8,000 records. Beauty has the largest number of records, at 11,809, while Home has the smallest, at 8,610. The categories are therefore all represented substantially in the dataset, although their record counts differ.

4.3 Monthly order activity

Monthly order counts fluctuate throughout the year. February has the lowest reported count, at approximately 3,800, while May has the highest, at approximately 4,300. The pattern does not show a consistent upward or downward trend across the year.

4.4 Relationships between numerical variables

The correlation heatmap shows an almost perfect positive correlation between Line_Total and Order_Total (approximately 1.00). Cost_Price and Selling_Price also have a strong positive correlation of approximately 0.96.

Line_Total and Order_Total have moderate positive correlations with Quantity_order (approximately 0.60) and stronger positive correlations with Unit_Price_order (approximately 0.73).

Return_Flag has little linear correlation with the other numerical features examined. This does not rule out nonlinear relationships or predictive value from combinations of features.

4.5 Customer segments

The boxplot comparing Bronze, Silver, and Gold segments shows considerable overlap in their order-total distributions. Each segment has a wide spread of transaction values and high-value outliers. The plot does not show a clear separation between segments based on order totals alone.

4.6 Return rates by category

The calculated return rates are:

Product category	Records	Returned records	Return rate
Home	8,610	442	5.13%
Beauty	11,809	600	5.08%
Fashion	9,260	447	4.83%
Electronics	9,969	479	4.80%
Groceries	10,352	474	4.58%

Home has the highest return rate at 5.13%, followed by Beauty at 5.08%. Groceries has the lowest rate at 4.58%. The difference between the highest and lowest rates is 0.55 percentage points, suggesting that the observed rates are relatively similar across categories.

These percentages are calculated at the dataset-record level. They should not automatically be interpreted as unique-order return rates unless each record is confirmed to represent a distinct order.

5. Business Implications

The findings suggest several areas for further investigation:

- **Transaction values:** Understanding the factors associated with high-value transactions may improve sales forecasting.
- **Category performance:** Differences in category record counts may help guide further investigation into product demand and inventory planning.
- **Return behavior:** Home and Beauty have slightly higher observed return rates. Investigating product-level and transaction-level factors may help explain these differences.
- **Customer segmentation:** The overlap in order totals suggests that customer segment alone may not clearly distinguish transaction values.
- **Feature relationships:** Highly related features should be reviewed before modelling to avoid redundant inputs and target leakage.

These are potential business opportunities rather than proven causal conclusions. The analysis describes patterns in the available data and does not establish why those patterns occur.

6. Conclusion

The ShopSmart Retail dataset provides a foundation for sales-value regression and return-flag classification. Exploratory analysis identified differences in category sizes, monthly order activity, transaction values, feature correlations, and category-level return rates.

The next stage is to develop and evaluate machine learning models using carefully selected features and appropriate preprocessing pipelines. Model performance should be assessed before using predictions to guide business decisions.